

Project Reorganization Complete

Date: May 5, 2026

Status: Complete & Tested

Changes Made

1. Python Scripts Reorganized

BEFORE		AFTER
phase2_langmuir_fitting.py	→	fluoride-adsorption-hybrid/src/
doe_design.py	→	phase1_doe_design.py
simulate_adsorption.py	→	phase1_simulate_adsorption.py

2. Data Organization

```
fluoride-adsorption-hybrid/data/
  raw/
    literature_data_extraction.csv    (Reference data)
  processed/
    doe_lhs_500.csv                  (LHS design)
    dataset_simulated_500.csv        (Simulated responses)
    dataset_simulated_500_CORRECTED.csv (Archive)
```

3. Results Structured by Phase

```
fluoride-adsorption-hybrid/results/
  phase2/
    langmuir_predictions.csv        Generated
    langmuir_model_info.json        Generated
    langmuir_diagnostics.png        Generated
  phase3/                           (For residual analysis)
  phase4/                           (For ML training)
```

4. Documentation Reorganized

```
docs/
  phase1/          (13 files) - Phase 1 research & factor analysis
  phase2/
    guides/        (2 files) - Theory & interpretation
    summaries/     (7 files) - Cheat sheets & execution guides
  project/         (5 files) - Roadmap, README, quick start
  references/      (4 files) - Literature & background
```

Code Changes

phase2_langmuir_fitting.py - Updated Paths

BEFORE:

```
DATA_FILE = 'data/dataset_simulated_500.csv'  
OUTPUT_DIR = 'results'
```

AFTER:

```
# Get script directory to support running from different locations  
SCRIPT_DIR = Path(__file__).parent.parent  
DATA_FILE = SCRIPT_DIR / 'data' / 'processed' / 'dataset_simulated_500.csv'  
OUTPUT_DIR = SCRIPT_DIR / 'results' / 'phase2'
```

Benefits: - Paths work from any directory - Paths auto-adjust to new structure - Tested & verified

Running Scripts

From Project Root:

```
cd /Users/ignite/College/IDP  
python3 fluoride-adsorption-hybrid/src/phase2_langmuir_fitting.py
```

From Project Directory:

```
cd /Users/ignite/College/IDP/fluoride-adsorption-hybrid  
python3 src/phase2_langmuir_fitting.py
```

Both methods work perfectly!

File Summary

Component	Before	After	Status
Python scripts	3 (scattered)	3 (in src/)	Organized
Data files	Mixed locations	raw/ + processed/	Clean
Results	results/ (root)	results/phase{2,3,4}/	Structured
Docs	20+ scattered	31 organized by phase	Clean
Empty dirs	data/, results/	Removed	Clean

Benefits of New Structure

Scalability - Easy to add Phase 3, 4, 5+ work

Clarity - Clear separation of concerns

Maintainability - Documentation organized by phase

Reproducibility - Scripts work from anywhere

Professional - Industry-standard project layout

Git-friendly - Ready for version control

Next Steps

Phase 3: Residual Analysis - Input: `fluoride-adsorption-hybrid/results/phase2/langmuir_predictions.c`
- Output: `fluoride-adsorption-hybrid/results/phase3/` - Script location: `fluoride-adsorption-hybrid/src/`
(create)

Phase 4: ML Model Training - Input: Residuals from Phase 3 - Output: `fluoride-adsorption-hybrid/results/`
- Scripts: RandomForest, XGBoost, MLP models

Notes

- All imports still work
 - Scripts tested
 - Path handling improved
 - Documentation preserved
 - Git repository intact
-

Status: Ready for Phase 3 work!